



An internet of things architecture for real-time tide and wave monitoring using Heltec LoRa 32 and Kalman filter

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Abstract

Tidal data monitoring is an important component for onshore and offshore activities. This study designs a low-cost tide gauge and an architecture that combines the Internet of Things with numerical processing to record tidal and ocean wave data from a single sensor without compromising data quality. The location of tide gauge test is located in Timbulsloko Village, Sayung District, Demak Regency, Central Java, Indonesia. The tide gauge test was carried out for 13 days from January 1, 2022, to January 14, 2022. This tide gauge study used a Heltec LoRa 32,433 MHz microcontroller. The A01NYUB type UART ultrasonic sensor is used to measure sea level and wave height. Tidal data is filtered using the Kalman method until the tidal and residual signals are obtained. Timbulsloko is a micro-tidal region, the tidal range is less than 2 m. The mean high water spring (MHWS) is 1.15 m while the mean low water spring (MLWS) is 0.15 m. The type of tide that occurs in the Timbulsloko area, Demak is a mixed tide prevailing diurnal. The mean wave height of the recorded residuals is 22.82 cm. This tide gauge system produces good quality data when compared to other studies that use standardized observation and instrumentation.

Keywords Tide gauge · Internet of Things · Kalman filter · Wind waves · Lora

1 Introduction

1.1 Coastal dynamic process

Understanding coastal processes such as morphological changes in the intertidal zone is an important study, because it can provide the first defence against waves and extreme sea level changes (Knight et al. 2021). There is a growing concern in global coastal zones about the costs of catastrophic flooding caused by storm surges or heavy rainfall

(Muis et al. 2016; Buchanan et al. 2017). Several studies have been conducted in various parts of the world and show a very high level of vulnerability faced by many coasts and deltas, especially in the intertropical and densely populated zones, such as Jakarta, Mumbai, and Bangkok (Nicholls et al. 2011; Hallegatte et al. 2013; Kulp and Strauss, 2019; Becker et al. 2020). Extreme sea level rise can worsen conditions along coastlines if changes in coastal morphology continue (Almar et al. 2021). The increase in greenhouse gases resulting from the burning of fossil fuels contributes greatly to the extreme warming of the climate system, thereby increasing the average temperature of this century and affecting ecosystems and humans (Oppenheimer et al. 2019). As a result of this warming, the polar ice caps melt and increase the volume of the oceans, causing global sea levels to rise and increasing the intensity and frequency of coastal disasters (Oppenheimer et al. 2019). Therefore, sea level measurements are very important as a response to global warming events and other extreme events in coastal areas (Cazenave et al. 2018; Marcos et al. 2019).

Changes in coastal conditions depend on various physical processes over a certain period of time and can originate from the sea, atmosphere, or land (Woodworth et al. 2019; Durand

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et al. 2019). The coast is a very dynamic area and is subject to change by the effects of waves, wind, and currents throughout the seasons. This change occurs due to the displacement of sediment and other changes in the coastal area due to extreme events such as storms (Knight et al. 2021). In various cases of vulnerability in coastal areas, there are important activities that need to be carried out by coastal engineers to monitor any changes on the coast using GPS, LiDAR, to satellites (Scott et al. 2016; Dealbera et al. 2021).

Tides and ocean waves are very important variables in the management, planning, and development of coastal areas. These two data serve as a reference in determining the location and height of coastal buildings such as harbors or breakwaters (Maulana et al. 2019). Tidal data monitoring is a very important component for onshore and offshore activities, especially for detecting spatio-temporal changes from global warming or other processes (Cazenave et al. 2018). Sea level measurement has long been an important component of many onshore and offshore services such as determining reference levels for national altitude systems, facilitating port operations and safety, and assisting navigation (Adebisi et al. 2021). In addition, sea level data is a reference in climate change analysis, such as the impact of sea level rise in coastal zones which is a major threat to more than 600 million people in the world (Neumann et al. 2015).

Tides are a phenomenon of periodic up and down sea-level movements caused by the movement of astronomical objects (sun and moon) (Ginanjari et al. 2019). Ocean waves can be generated by wind, earthquakes, to gravity. Wind-generated waves generally have shorter periods ranging from <0.1 s for capillary waves and a maximum period of 30 s for the longest swell waves in the open ocean. These waves are a combination of long-period (10–20 s) large waves and short period (8–10 s) local wind waves with periods varying between 8 and 20 s (O'Donncha et al. 2019).

1.2 Measurement of sea level

Presently, Internet of Things (IoT) is widely used for sea level measurement. Internet of Things is a paradigm that possibly use for communicating between electronic devices and sensors through internet connection (Kumar et al. 2019). Apart from the measurement concept of sea level using IoT, many research related to tide gauges has been carried out by various researcher around the world. Sugianto et al. (2022) carried out water level measurements using ultrasonic sensors but had not used the internet of things method to analyze erosion in the Timbulloko area. Haq et al. (2021) made a low-cost tide gauge using the Internet of Things concept with an Arduino microcontroller and an ultrasonic sensor of the JSN SR-04T type. The tide gauge used data transmission at intervals of 15–19 s and there was a delay due to internet access via mobile hotspot. Haq et al. (2021) also applied

moving average and moving media filter methods for filtering tidal data. The research has not implemented the Long Range system under certain conditions. Knights et al. (2021) conducted a low-cost tide gauge experiment in Liverpool using an Arduino Sigfox MKR 1200 microcontroller and a Blue Robotics 2-bar Pressure Sensor to measure water level. This study uses the concept of hydrostatic pressure which will measure changes in water level through changes in seabed pressure and data transmission has used the Long Range (LoRa) system. The resulting data is very smooth and there is no ocean wave noise. Prasetyawan et al. (2021) tested a tide gauge named PUMMA (Inexpensive Measurement Tool for Water Level). PUMMA uses ultrasonic sensors and this system is placed in the vegetation area. This system is equipped with CCTV to monitor the condition of the ecosystem. However, PUMMA has a fairly long data gap and affects tidal analysis for navigation purposes.

In addition to using tide gauges, measurement of water level can use altimetry satellites. Variations in water level can be seen with altimetry satellites, including thermal, glacier, continental freshwater fluxes, and others. These observations are available on a variety of platforms: services that manage Archiving, Validation and Interpretation of Satellite Oceanographic data (AVISO), Climate Change Initiative (CCI), Commonwealth Scientific and Industrial Research Organization (CSIRO), University of Colorado (CU), and Center for Space Flight Goddard (GSFC) (Dealbera et al. 2021). Tidal data recording can be done using satellites or tidal stations. Altimetry satellites have provided sea surface height (SSH) dynamics data since 1993 (Saraceno et al. 2008). However, the dynamics of the coast make satellite altimetry not sufficient to complete coastal processes. These problems include the intrinsic difficulty of influencing the correction applied to the nearshore altimeter data and the difficulty of interpolating the two satellites orbiting on their respective paths (Le Traon et al. 2002). This problem makes the tidal data used in many coastal models not accurate enough, due to the large resolution of satellite imagery and not detailed temporal recordings. Therefore, tidal stations are needed to record the water level more accurately and continuously from time to time. Tidal logging in tidal stations usually ignores wind-wave noise due to the tighter timing of the recording resolution. In fact, this wave data is very important and will be very useful in planning the development of coastal areas. In addition, tidal data recorded by tidal stations do not represent many areas that are quite strategic and important. A tide gauge in Indonesia has a very expensive price, ranging from 2100 to 6650 USD, which makes many researchers choose to use tide staff (Kurniawan et al. 2016).

1.3 Measurement of ocean waves

As for ocean waves, there are several studies that use different instruments for measurement observations. Wave measurement technology has developed over the last 40 years (Pandian et al. 2010). Ginanjar et al. (2021) used the JL-FS2 anemometer to measure the significant wave height (H_s) and significant wave period (T_s). In this study, the wind speed in knots was converted to the height and period of significant waves directly using the wind-wave equation. This measurement has been integrated with the Internet of Things (IoT) and uses the Arduino UNO R3 microcontroller and the NodeMCU ESP8266. Sugianto et al. (2022) used the ADCP (Acoustic Doppler Current Profiler) instrument to measure wave height in Timbulloko waters. Measurements using this instrument are quite accurate but require quite expensive costs due to the relatively expensive price of the instrument. Munandar et al. (2018) made a wave measurement instrument using the buoy concept in the waters of Palabuhanratu, Indonesia. Buoys are used worldwide to measure surface waves, namely as frequency or validation of wave forecasts and models (Krogstad et al. 1999; Li and Saulter, 2012; Christensen et al. 2013). In this study, an accelerometer sensor type ADXL 345 was used to measure speed and used Arduino Mega 2560 as the microcontroller. The accelerometer sensor uses the principle of position displacement as acceleration based on the Cartesian coordinate system (XYZ). The data storage system of this study uses an SD Card and has not used the Internet of Things system.

Based on these conditions, this research designs a multifunctional that has a low price and an architecture that combines the Internet of Things with numerical processing to record tidal and ocean wave data simultaneously from one sensor without compromising the data quality of each component. The advantage of this tidal meter is that it can send, store, and display data automatically and can be accessed quickly and in real-time (Husrin et al. 2021). The development of tide gauges with the Internet of Things architecture will be easier to install and less complicated maintenance, so it is very supportive of monitoring coastal conditions. The development of this tide gauge is very much needed at the research location, namely on the coast of Demak, Timbulloko. This is because Timbulloko is an area with severe erosion and sedimentation conditions (Bijsterveldt et al. 2020). To mitigate the abrasion and sedimentation that occurs, data such as tidal data and wave characteristic data are needed. The development of a tide gauge on the coast of Demak, Timbulloko is needed to obtain these data. These systems can fill technology gaps and contribute to larger projects such as Smart Cities (Kim et al. 2017) and Smart Coastlines (Bresnahan et al. 2020). A tide gauge system that does not ignore ocean wave noise data has the advantage of identifying significant coastal morphological changes such

as sediment transport and storm waves. In addition, a tide gauge network like this can complement other regional tide gauges as well as for port navigation purposes (Metters et al. 2017) as well as disaster mitigation such as tsunamis and storm surges (Joseph and Prabhudesai 2006). In the IoT system in Galway Bay, Ireland, data such as weather, water, waves, tides, currents, seafloor mapping, and topography are collected from various sources and used to inform industry (Dlodlo et al. 2012). Meanwhile, controlling the shipping industry at the port requires real-time information about the tides and speed of ocean currents.

2 Methodology

The location of the tide gauge test is located in Timbulloko Village, Sayung District, Demak Regency, Central Java, Indonesia (Fig. 1). Tide gauge testing was carried out for 13 days on January 1, 2022 at 12:00 to January 14, 2022 at 23:59 in western Indonesia time with a test interval of 1 min to obtain detailed tides during spring tides and neap tides and sea waves caused by wind. The research area is 13.4 km from the government-owned tidal station located in Semarang.

Seasonal conditions in January for the Timbulloko region are the West Monsoon or Asian Summer Monsoon. The Asian Summer Monsoon is characterized by a northwesterly wind blowing from Asia to Australia, which brings moist air to cause the rainy season in most parts of Indonesia (Alifdini et al. 2021; Griffiths et al. 2009; Chang et al. 2005, 2006). This seasonal system makes the characteristics of the Timbulloko area which is directly opposite the zone of wind flow from the Java Sea resulting in very large waves in January in the Java Sea and on the coast of Timbulloko there will be oscillating wind waves. Under certain conditions, there can be quite high atmospheric disturbances in the region due to the low pressure that occurs at the equator. This can cause high wind gusts and big waves. In addition, the intertidal zone

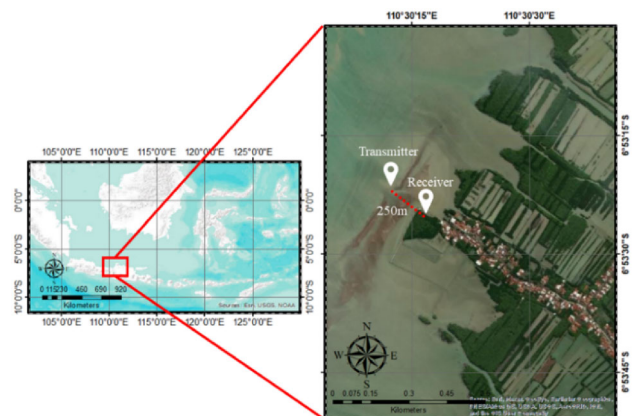
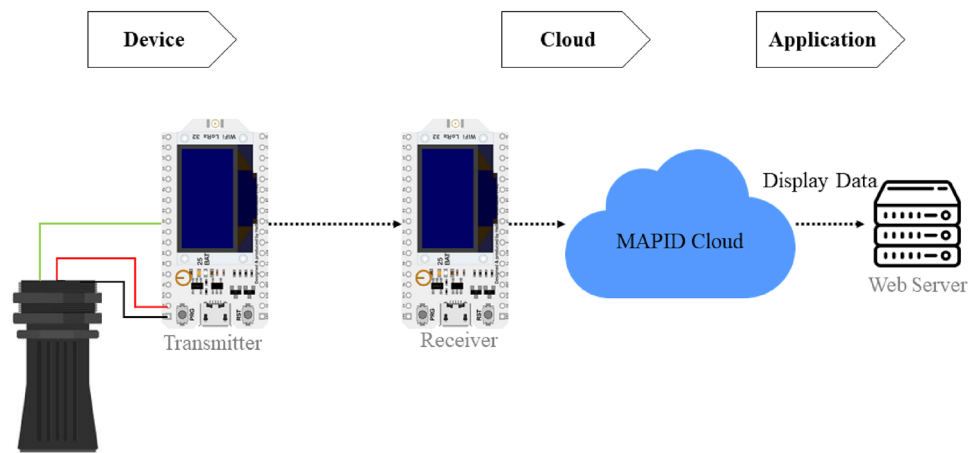


Fig. 1 Research location

Fig. 2 Internet of things for tide gauge system architecture



in the Timbulsloko area has a low current velocity and the types of sediment are silt and clay (Utomo et al. 2022). The silt-type sediment has a cohesive nature where it is difficult to be distributed by currents (Rahmadi et al. 2021).

2.1 IoT tide gauge network design

The IoT concept is widely used in research, especially in disaster management (Ghapar et al. 2018) to marine applications. The term IoT was coined by Kevin Ashton in 2009 and he defined IoT as a connected object that can be uniquely identified using radio frequency identification (RFID) technology (Ashton 2009). Currently, there are many tidal measuring networks around the world, eg national networks covering all coastlines and local networks for internal port purposes (Knight et al. 2021). These systems use various communication methods for sending data to users (Holgate et al. 2008; Perez et al. 2013). The tide gauge network is very much needed for the purpose of monitoring the accurate sea level on the coast. These networks require low voltage consumption for long-term use. In addition, multi-station can be applied using a gateway, so that it can record data and send it simultaneously to the same central device. The tide gauge network must have accurate records and real-time data transmission for beach monitoring. The tide gauge designed here is specifically for recording tidal and ocean wave data simultaneously, then filtered using a Kalman filter. This system can be adapted to fill gaps in tidal and ocean wave data that occur in an area (Fig. 2).

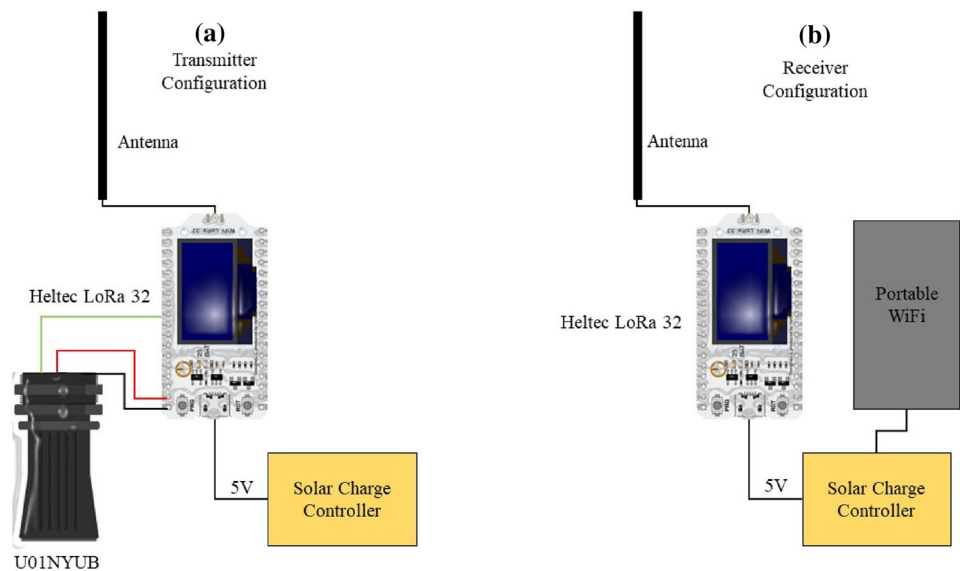
This tide gauge study uses Heltec LoRa 32 433 MHz microcontroller which is equipped with the same frequency antenna. This microcontroller uses the SX1276/SX1278 chip and can support node-to-node or LoRaWAN communication. In this study, Heltec LoRa 32 uses node-to-node communication, so it requires two LoRas that act as transmitters and receivers to transmit tidal data. This microcontroller has a maximum output power of 19 dB with fairly low power con-

sumption. When using a 12 dB LoRa output, the current required is about 60 mA. Heltec LoRa 32 works on a voltage of 4.7–6 V using power from USB. Berto et al. (2021) did a distance experiment using Heltec LoRa 32 with several scenarios. Heltec LoRa has efficiency in the Broadband adjustment (250 kHz) less than 50% after transmitted data with distance in 800 m, however, it has a quick sending time, which ~500 ms. Other than that, the best scenario is using the NarrowBand + adjustment (125 kHz) with efficiency 90% in so far as 1600 m with transmitted time ~5.3 s. This microcontroller was chosen, because it has a fairly cheap price and can be used to transmit data with Long-Range or an internet connection, so the use of tide gauges is very flexible.

The ultrasonic sensor to measure sea level and wave height uses a sensor type with an A01NYUB type UART output with a maximum detection of 750 cm and a resolution of 1 mm. This sensor has a fairly fast response time, which is 100 ms, and a very low current consumption, which is less than 15 mA. In addition, the blind zone on this sensor is not too high at 0–28 cm, so it can still be used as a tide gauge with a maximum tidal range of 7 m from maximum detection. The power consumption of this sensor is around 3–5 V, this consumption is normal for an ultrasonic sensor for a detection range that is not too high. The UART output on this ultrasonic sensor can be processed by Heltec LoRa 32, because the microcontroller has the ability to read sensors with UART output.

The Internet of Things architecture in this research tide gauge uses two data transmission systems, the first system transmits water level data and ocean wave noise recorded by ultrasonic sensors using the Long Range (LoRa) system. This is done because there is no strong internet signal in the observation area, so data transmission using a Long Range system is needed. At this stage, data are sent from the transmitter and received by a receiver with the same type of microcontroller, namely Heltec LoRa 32, 433 MHz. In the second system, the data received by the receiver is then sent to a database with an

Fig. 3 Configuration system
a transmitter b receiver



internet system. On the receiver side, the internet condition is quite good and it is possible to send data to the database. The database used is the one available on the Internet of Things MAPID facility (<https://console.mapid.io/>). Data sent to the MAPID database are stored in a comma-separated values (CSV) format. The Long Range system on this tide gauge is node-to-node, where the receiver receives data from a single transmitter source (Fig. 3).

The transmitter system uses a power supply from a 50wp solar panel and two 12 V 20Ah accumulators installed in parallel. To regulate excess power and supply power to the microcontroller, the transmitter system uses a solar charge controller. While in the receiver system, the data supply uses two 50 wp solar panels and two 12 V 20Ah accumulators which are installed in parallel. The receiver system also uses a solar charge controller to regulate the overvoltage. In the receiver system, there is portable internet to send data to the MAPID cloud and display it on a web server. The amount fund to create one tide gauge, with one transmitter and one receiver, is 412 USD.

2.2 Kalman filter

Kalman filter is widely applied in signal processing and instrumentation. Kalman filter often use to identify time series model, localization and to track moving target, denoising signal, deconvolution, network sensor wireless and estimate distribution. Instrumentation is obviously one of the main function to applied Kalman filter. The advantages of using Kalman filter are mainly in sensor fusion, sensor calibration, ultrasonic estimation, frequency measurement, GPS, and more (Auger et al. 2013).

In the transmitter system, recorded sea-level elevation data are separated into two main data, namely raw data and

tidal data without noise. Tidal data without noise were filtered using the Kalman method. The Kalman method is a mathematical method that provides a filter solution using the least-squares method (Fragher 2012). Kalman's algorithm is applied to a system that is represented in a state of space and the operation is basically divided into two, namely prediction and upgrade (Guerrero et al. 2015) (Fig. 4).

Information:

Prediction:

Priori estimate:

xt = State vector containing the data of interest.

ut = Vector containing any control inputs

Ft = State transition matrix which applied the effect each system state parameter at time $t - 1$

Bt = Control input matrix which applied the effect each control input parameter in the vector ut

wt = Containing process noise terms for each parameter in the state vector

Error covariance:

Pt = Error covariance

Qt = Process noise covariance matrix associated with noisy control inputs.

Correction:

Kalman calculation gain:

Kt = Kalman gain

Ht = Matrix showing the relationship between measurements and the state vector

Rt = Covariance matrix (measurement noise)

Estimate update:

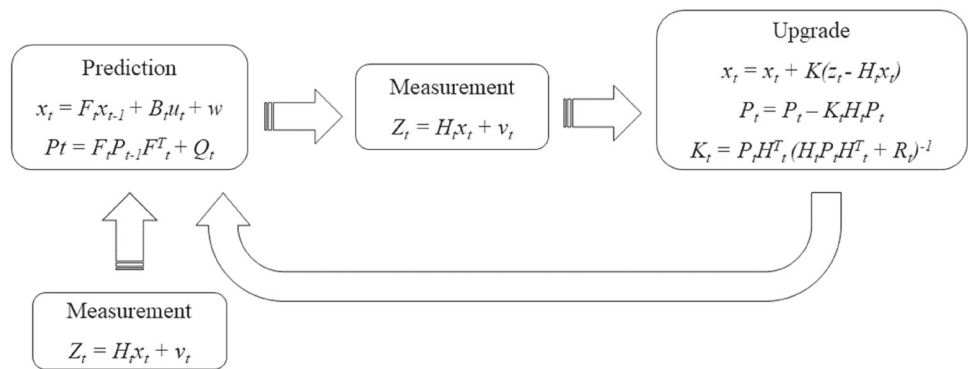
Error covariance update:

Measurement:

Zt = Measurement vector

vt = White noise

Fig. 4 Kalman filter



2.3 Separation of wave data

Separation of tidal data noise was done by reducing the value of tidal data that had noise with tidal data from the Kalman filter. The residue from the reduction was data on ocean waves generated by the wind.

$$\text{Wave} = \text{Data}_{\text{noise}} - \text{Data}_{\text{Kalman}} \quad (1)$$

$\text{data}_{\text{Noise}}$ = tidal data that has noise, $\text{data}_{\text{Kalman}}$ = tidal data that has been carried out by the Kalman filter.

2.4 Spectral analysis

According to Pan et al. (2018), the wavelet transform is basically a Fourier transform where the window on the wavelet transform can be adjusted by following the following equation.

$$W(a, b, x, \psi) = \int_{-\infty}^{\infty} x(t) \psi^* \left(\frac{tb}{a} \right) dt \quad (2)$$

where $x(t)$ is the input time series, ψ^* is the basic wavelet function, a is the frequency scale, b is the original translation. Physical interpretation of the equation is very simple, namely $W(a, b, x, \psi)$ is the 'energy' x of scale a and time b (Huang et al. 1998). For flexibility, we used continuous wavelet transform (CWT) for tidal analysis (Jay and Flinchem, 1997; Pan et al. 2018). In this study, the Morlet type of wavelet function is used because it is optimal for tides (Sassi and Hoitink 2013; Guo et al. 2015; Hoitink and Jay, 2016; Losada et al. 2017; Pan et al. 2018).

The CWT of time series data is basically a representation of the data in different database vector sets. Its individual components are no more reliable estimates than a single data point viewed separately. Yet, it is these individual estimates that are needed to analyze the events that make up the non-stationary record in tidal data (Flichen and Jay 2000). In the study of ocean engineering, sea conditions such as significant wave height, wind, and sea level characterize sea conditions

in the time domain where the process is assumed to be stationary. In general, this assumption will apply if the wave period is not more than 1 h (Solari and Losada, 2011). Therefore, that in this study, there is a wave range with a period of about 1 min to 24 h which can be assumed as stationary or non-stationary data. For example, a wave period of 5 min to 12 h which is a type of long-period wave and is influenced by wind conditions and other meteorological conditions can be assumed to be non-stationary.

3 Experimental setup

Tide gauge transmitter is made as high as 2.5 m from the seabed and installed using concrete on the seabed. Then, the ultrasonic tide gauge sensor on the transmitter is installed as high as 1.96 m from the seabed. The LoRa antenna is placed 20 cm above the tide gauge building and has a lower position than the receiver antenna which is installed at a height of 5 m above the seabed. The distance between the antennas from the transmitter to the receiver is about 250 m. The antenna on each LoRa is mounted vertically. Sensor readings on the transmitter are set every one second to be sent to the receiver. At the receiver, the received data are then filtered using a Kalman filter with Measurement Uncertainty of 20 data, Estimation Uncertainty of 2 data, and Process Noise of 0.01. Data are sent to the database every 1 min, so the data per second sent from the transmitter will be smoother. Then, we compared the recorded data with comparison test using standardized tidal data from tidal station owned by the Geospatial Information Agency (BIG). The distance between transmitter tide gauge with the nearest tidal station owned by the Geospatial Information Agency (BIG) is 13 km, which located in Semarang, Central Java, Indonesia (Figs. 5, 6).

Timbulsloko is a micro-tidal region, the tidal range is less than 2 m. The mean high water spring (MHWS) is 1.15 m while the mean low water spring (MLWS) is 0.15 m. The type of tide that occurs in the Timbulsloko area, Demak is a mixed tide prevailing diurnal (Qarnain et al. 2014; Utomo et al. 2022). This can be seen visually in the graph of the mea-

Fig. 5 The tide gauge transmitter system

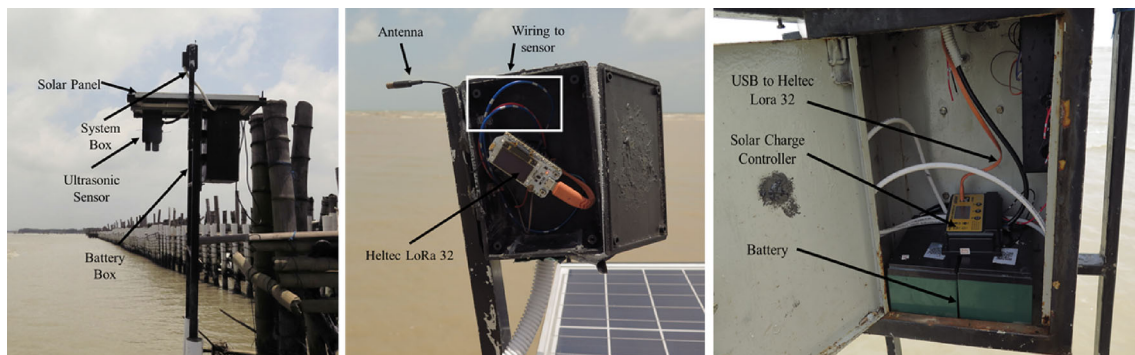
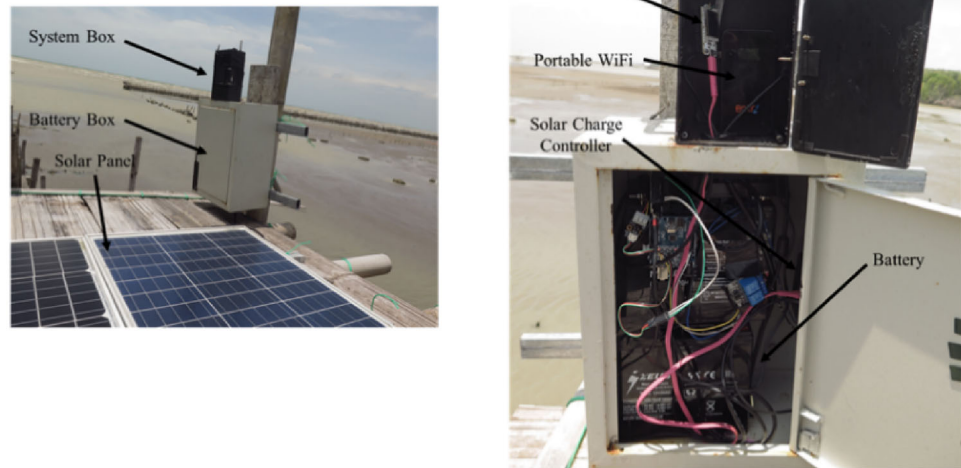


Fig. 6 Receiver tide gauge system

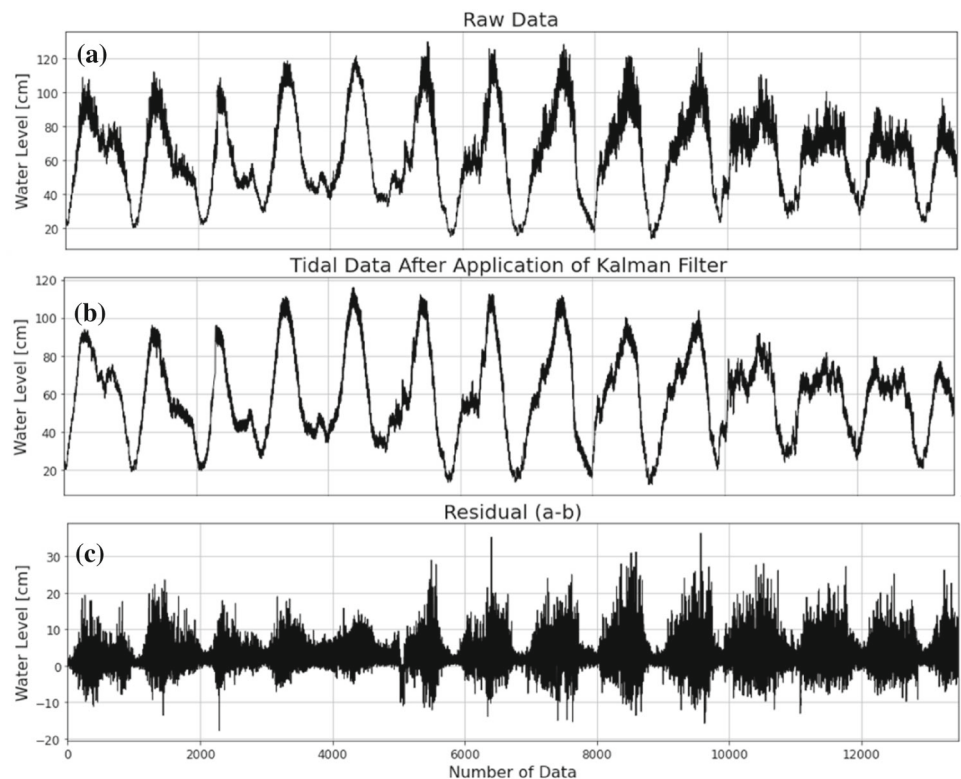
surement data which shows a mixed tide prevailing diurnal and there are several mixed days. In addition, the daily sea waves that occur in the Timbulsloko area are not only influenced by seasonal winds, but also by sea and land breeze conditions as well as bathymetry, so it can be seen that the wave height will be dominant during high tide and at night. The results of tidal recording using this tide gauge system get the tidal signal and its noise.

4 Result analysis

In Fig. 7a shown are tidal data that oscillates according to the type of tide and there are other waves that oscillate with shorter periods. These shorter oscillating waves are categorized as wind waves because they have a shorter period of ebb and flow. Tidal data in Fig. 7a. This is raw data that requires a filter so that it gets separate tidal and other wave data. The results of recording from the sensor there is no sig-

nificant error data. The filter results from this data are shown in Fig. 7b. The graph is the result of the Kalman filter which shows the data is smoother and there is only a small amount of recorded noise waves. The filter using the Kalman method does not reduce the quality of the tidal data produced, so it can be seen that the recorded tidal pattern remains the same as raw. On the other hand, the resulting output data will be more accurate when processed with Admiralty or Least Square to get the harmonic components. The results of filtering data using Kalman will produce harmonic components that correspond to long-period waves that occur at high tide without the occurrence of waves with shorter periods. This is very beneficial when the desired data is only tidal data without removing the noise that occurs in tidal data, so the Kalman filter can be applied in the manufacture of tidal systems using ultrasonic sensors. The residue from the Kalman filter will be a residue that can be classified in the form of a short-period waveform as shown in Fig. 7c.

Fig. 7 Tidal and wave graphs
a raw data, **b** tidal data after application of Kalman filter,
c residual (**a**, **b**)



Results in Fig. 7c show oscillating waves with a shorter period than the tidal data. In 1 day, there are hundreds or even thousands of oscillations caused by the transformation of wind energy that blows on the sea surface into wave energy. This oscillation varies from 0.1 s to 5 min, so when recording data with a period of 1 min, it will get a range of shortwave data with a period between 1 and 5 min. Waves with this period are categorized as infragravity waves which are the result of wind forces and atmospheric pressure gradients. Ginanjar et al. (2021) observed that the waves that occurred in Timbulsloko using wind parameters had an average height of 46 cm in January. While Hawati et al. (2017) observed that waves in the Timbulsloko area in January using observation data with the Sontek Argonaut Type Xr 750 MHz Wave Recorder instrument had a significant height (H_s) between 24.66 and 30.32 cm. The range of time series wave data from these two studies is set not to exceed zero, so when the residual tide gauge results are set the same, it will produce an average wave value that is almost the same as the two studies, which is 20.98 cm. Based on this, it can be seen that the results of the tidal residue are short-period waves which can be classified as wind waves or infragravity waves. However, the resulting data are not a significant wave height (H_s).

The results of observations of tidal stations on the system that have been built are then compared with the data recorded from fixed tidal stations owned by the Geospatial Information Agency (BIG) in the Semarang area. Because the distance between the two stations is about 13 km, the

tidal range recorded at the two stations is different. Therefore, each data are filtered with its respective MSL value, so that the MSL value in both data is equal to zero. This is done to see the similarity of the pattern and range of oscillations in the two data. In Fig. 8, it can be seen that the pattern between the data recorded by the BIG fixed tide station and the results of system observations has fairly high similarity. Likewise, the value of the tidal oscillation range is not much different. On the other hand, it can be seen that the tide gauge system has very high stability, which is indicated by the lack of outliers in the observations. Therefore, based on this, it can be ascertained that the sensors used in this tide gauge system have very good accuracy.

In Fig. 9 shows a relationship between BIG data and system recording. The relationship between the two results in the value of $RMSE = 15.229$ and $Bias = 4 \times 10^{-14}$. These two values show that the quality of tide gauge system data has a very good quality and approach the actual condition which recorded by fixed tide station from BIG. The tide gauge system has a high accuracy, it has the same pattern which so similar to the actual condition with fixed tide station.

Figure 10 shows the results of spectral analysis of tidal data from BIG and sensor measurement using Fast Fourier Transform (FFT). Based on the graph, there are several waves with periods shorter and longer than the tides recorded in the tidal dataset recorded by the system. The shortest period waves that can be recorded are infragravity waves with a normal period of 30 s to 5 min (Herbers et al. 1995). However,

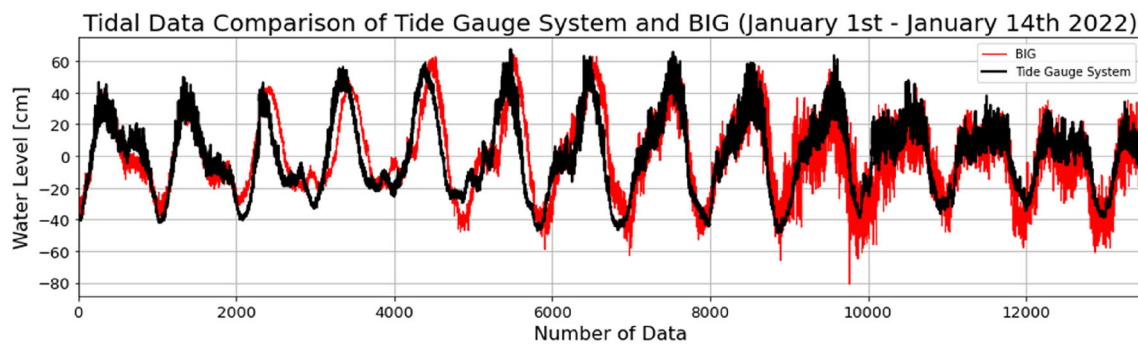


Fig. 8 Tidal comparison between BIG data and Tide gauge system observation

Fig. 9 Scatter plot BIG data and Tide gauge system observation

Scatter Plot Tidal Data of Tide Gauge System and BIG (January 1st - January 14th 2022)

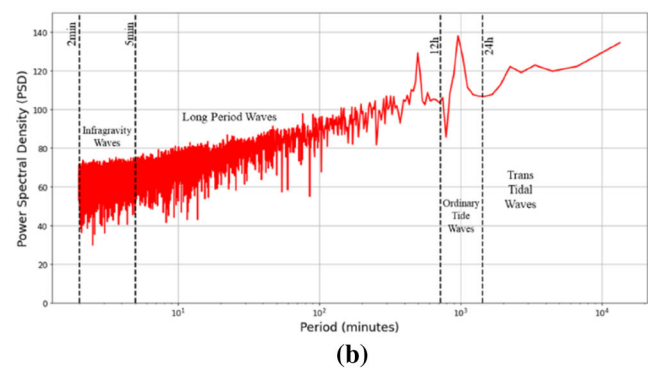
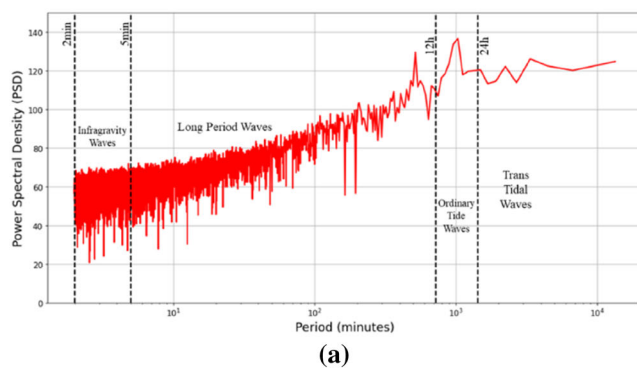
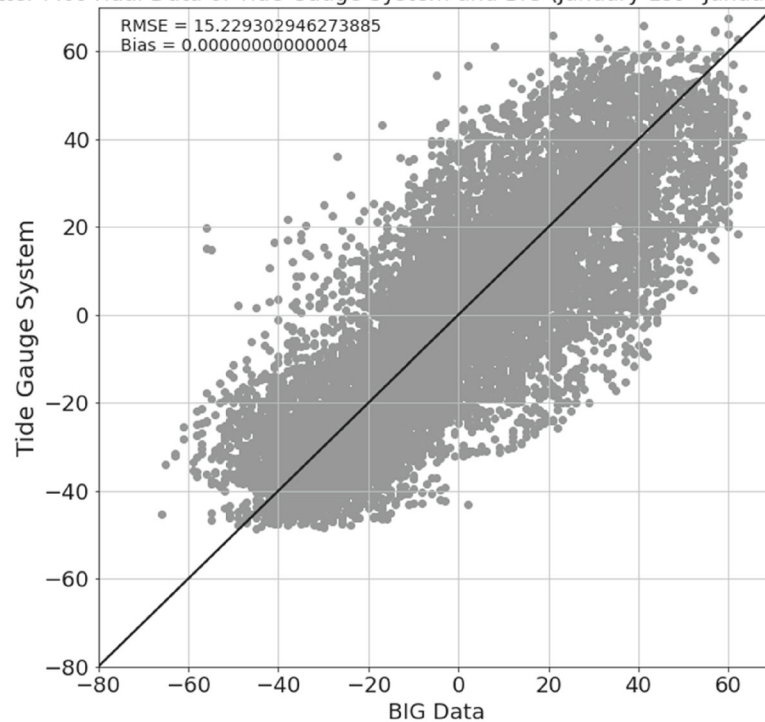


Fig. 10 Fast Fourier transform tidal wave data **a** tide gauge system, **b** BIG data

the shortest period of wave is wind wave which divides into ordinary gravity waves, ultra-gravity waves, and capillary waves that have period of 1–30 s. In coastal areas, incoming sea breezes and large waves dissipate their energy in the form of depth-induced breaking, releasing a less energetic component of infragravity (Toffoli et al. 2017). In general, infragravity wave motion can be found on the continental shelf at a depth of 8–200 m which is a subharmonic mixture that is limited by incident waves (Herbers et al. 1994). Based on the results of the Fast Fourier Transform, the lowest period of infragravity waves is 2.88 min and the longest is 5 min. Then the period between 5 min and 12 h can be recognized as long period waves. These waves are waves generated by meteorological phenomena such as weather disturbances or wind variations (Toffoli et al. 2017). Furthermore, the wave period between 12 and 24 h is a tidal wave produced by the movement of the moon and sun. This wave energy is dominated by semi-diurnal or diurnal types. In addition, tidal transe waves are also recorded in the observation system. This wave has a period of more than 24 h and is included in the wave with the longest period. In total there are four types of waves recorded by the system based on the results of the Fast Fourier Transform, namely infragravity waves, long-period waves, ordinary tidal waves, and trans-tidal waves. These waves are well represented in the Fast Fourier Transform graph and can be identified by their period of oscillation.

The comparison of the two data using FFT shows the same distribution of waves, namely infragravity waves, long period waves, ordinary tide waves, and trans tidal waves. The shortest period recorded by the sensor is 2 min in infragravity waves. This happens because the average data measurement interval recorded by the sensor is 2 min. So that the data recorded by BIG also adjust to the condition of the data recorded by the sensor to see the similarity of the patterns that occur in the FFT results. It can be seen that after processing with FFT, there are similarities in wave patterns and power spectral density (PSD) that occur in both data.

Based on the spectral graph using the Fast Fourier Transform, it is known that the ocean waves recorded by the IoT system vary greatly with the recording period of 1 min. This shows that the residual wave generated from the tidal data and the Kalman filter is a wind wave that has a period below the tide. This statement is reinforced by the results of the spectrum shown in Fig. 11. This spectrum is the result of processing using a Continuous Wavelet Transform (CWT) which transforms the signal from tides into a spectrum with a time–frequency domain. There are two spectrums which the result from the BIG data and tide gauge system. Two of them shows that are many variative classification of wave periods. Based on this spectrum, there are three main parts that dominate the tidal signal, namely the tidal wave which has a period of 0.5–1 day. This tidal wave in the CWT spectrum has a very dominant strength in these waters, so it can be seen

that the tidal wave in the Timbulsloko area is the main wave that oscillates according to the period of oscillation. Then the waves under a period of 12 h to 5 min are long period waves resulting from the process of wind variations and weather disturbances in the observation area. This is evidenced by the spectrum in the long period waves that have random energy in the spectrum column. This random energy is caused by changing atmospheric conditions interacting with the ocean and occurring locally in the Timbulsloko area. This wave is quite strong in the period from 1.5 h to 12 h. In addition, the role that has a fairly large dominance is infragravity waves with an oscillation period of 2.88 min to 5.6 min which is depicted in the CWT spectrum in Fig. 11. In the spectrum of infragravity waves, there are day and night effects that occur during high tides and low tides. In the Timbulsloko area, high tide occurs at night and low tide occurs during the day, so the dominant wind-wave occurs at night. This makes the CWT spectrum have a day and night pattern due to the tidal processes that occur in the area. In contrast to the spectrum of infragravity waves that occur in the dataset in the Timbulsloko area, the infragravity waves recorded in the BIG dataset show irregularities in these waves. This is due to the location of the tidal station which is not influenced by coastal conditions such as river estuary flows and land and sea breezes. Even though they are different, the identification of the two shows almost the same wave division.

Based on the filter results and the patterns found in the FFT and CWT, it can be seen that the noise at the tide has an important component in the form of waves that have a shorter period than the tides. These waves can be utilized in the process of coastal development, conservation, industry, and disaster mitigation such as storm surges. The waves with short-period siphons can be recorded properly by the IoT system in this study, so it is very clear that the difference between each wave is in the frequency domain and spectrum results.

5 Conclusion

In this study, we designed a tide gauge with the Internet of Things architecture and numerical processing to record tide and ocean wave data from one sensor without compromising data quality. Experiments were carried out with the distance between the transmitter and receiver as far as 250 m and data recording at 1-min intervals. Based on the experimental results, it is known that the tidal type in Timbulsloko is a mixed tide prevailing diurnal with the Mean high-water spring (MHWS) is 1.15 m while the mean low water spring (MLWS) is 0.15 m, and the mean wave height of the residuals recorded is 22.82 cm. These results are obtained from the tide gauge recording data filter using the Kalman filter. This tide gauge system produces quite good data quality when com-

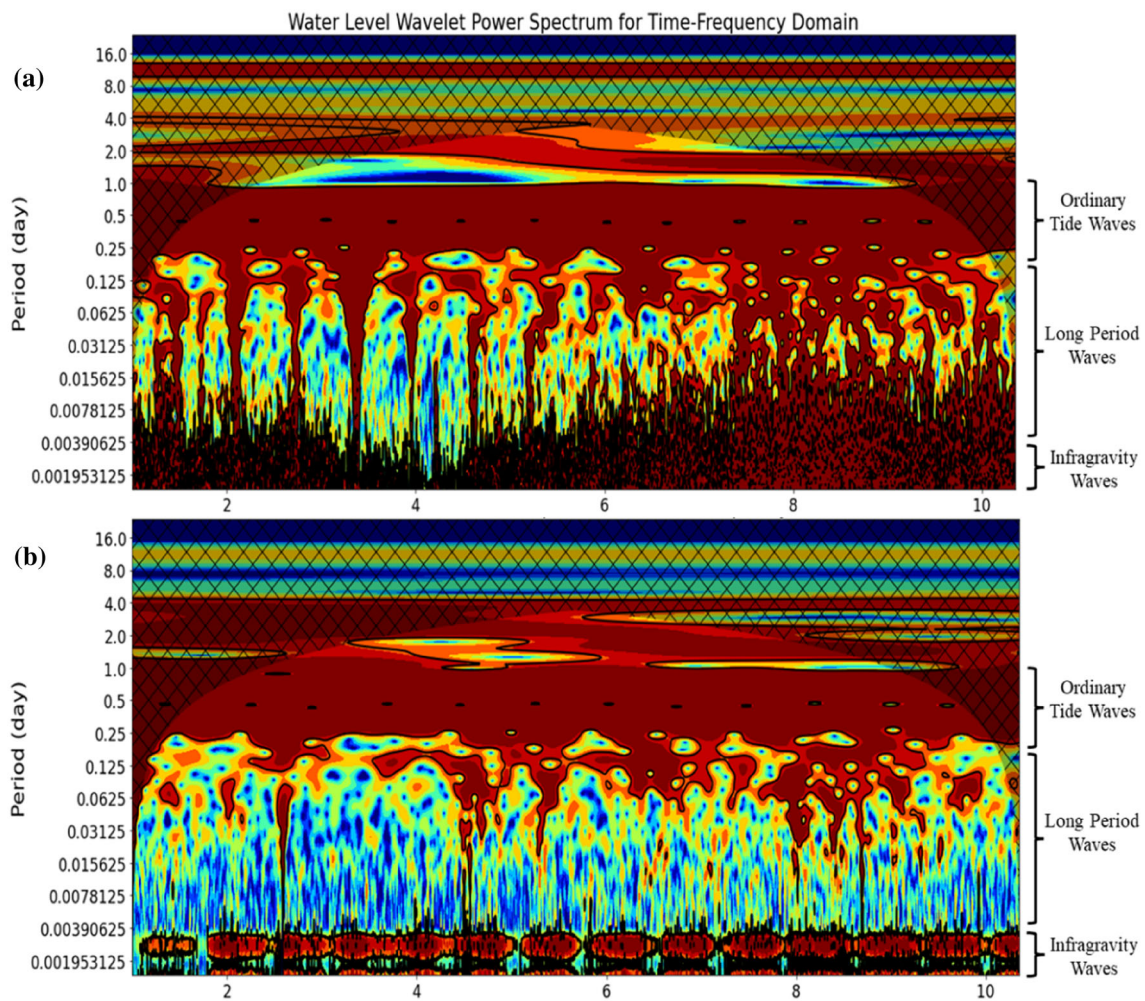


Fig. 11 Continuous wavelet transform tidal wave data **a** BIG data, **b** Tide gauge system

pared to other studies that use standardized observation and instrumentation. The results of the transformation using Fast Fourier Transform (FFT) and Continuous Wavelet Transform (CWT) show variations in short-period waves that occur in the Timbulsloko area based on the IoT system. Thus, the data provided by our developed monitoring system can be benefit for the mitigation efforts of the erosion and sedimentation in the coastal area of Timbulsloko. The transmission of data from the transmitter to the receiver is very good and only a little sensor noise other than wind waves is recorded in the data. This research can be used as a reference for developing tide gauge instrumentation for port and industrial activities, in future research it can add noise parameters that occur due to storms, rain, or low-pressure centers that occur around the observation site. In addition, can use other better filter methods.

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